Skills Summary

Circular Reasoning and Unsupported Statements

Use this reference while completing your worksheet or when studying.

Skill 1 — When a reason just repeats the claim

Test every reason with one question: was this fact available *before* this step? A reason may be a given, a definition, a postulate, or a theorem already proved. If the reason is the statement itself — or the theorem you were asked to prove — the step is **circular** and proves nothing.

Example: Prove vertical angles congruent. A student writes: (1) the angles are vertical — Given; (2) their measures are equal — vertical angles are congruent; (3) the angles are congruent — definition of congruence.
Step 1. Read each reason and ask whether it was available: step 2 cites the very theorem being proved, so the proof assumes its conclusion.
Step 2. Repair it with facts that come earlier: each angle is supplementary to the same third angle (two linear pairs), and angles supplementary to the same angle are congruent.

Try it: "$\overline{AB} \cong \overline{CD}$ because they are congruent segments." Which rule does that reason break?

check: it restates the claim — cite the given or the definition of congruent segments

Skill 2 — Use only what the givens support

Watch the words you were handed. "Linear pair" gives you supplementary ($+ = 180°$). "Vertical angles" gives you congruent ($=$). "Corresponding angles" gives you congruent only when parallel lines are given. Picking the wrong one changes the equation you solve, so it changes the answer.

Example: $\angle PQR$ and $\angle RQS$ form a linear pair, $m\angle PQR = 4x + 10$ and $m\angle RQS = 6x - 20$. Find x.
Step 1. The given is a linear pair, which supports supplementary — so add the measures to 180, do not set them equal.
Step 2. $(4x + 10) + (6x - 20) = 180$ gives $10x - 10 = 180$, so $10x = 190$ and $\boldsymbol{x = 19}$.

Try it: Two angles forming a linear pair measure $3x + 5$ and $7x + 15$. Find x.

check: $\boldsymbol{x = 16}$

Skill 3 — Name the postulate that does the work

Segment Addition Postulate: if M is between A and B, then $AM + MB = AB$. Betweenness alone gives you this. **Midpoint** is a stronger, separate given: only then may you write $AM = MB$.

Example: M is between A and B with $AM = 3x - 1$, $MB = x + 5$, and $AB = 28$. Find x.
Step 1. The given is betweenness, not a midpoint, so the postulate that applies is segment addition — add the parts to the whole.
Step 2. $(3x - 1) + (x + 5) = 28$ gives $4x + 4 = 28$, so $4x = 24$ and $\boldsymbol{x = 6}$.

Try it: M is between A and B with $AM = 2x + 1$, $MB = 3x - 6$, and $AB = 30$. Find x.

check: $\boldsymbol{x = 7}$